

① Existence of inertial reference frames.

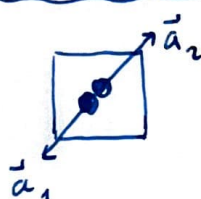
(attached coordinates axes where we make our measurements)

② $\vec{F} = \frac{d}{dt}[\vec{p}]$, $\vec{p} = m\vec{v}$ $\longleftrightarrow$ $\vec{F} = m\vec{a}$
 constant mass

in opposite directions

③ Action-reaction for central forces.

Measuring mass: If 2 bodies form an ideal isolated system, then accelerations of them are always opposite in direction.
Total force in an isolated system: $\sum_i \vec{F}_i = 0$ (think of an inflated balloon.)



during collision

$$\vec{F}_1 + \vec{F}_2 = 0 \rightarrow \vec{F}_1 = -\vec{F}_2$$

$$\Rightarrow \frac{d}{dt}[\vec{p}_1] = -\frac{d}{dt}[\vec{p}_2] \Rightarrow m_1 \frac{d\vec{v}_1}{dt} = -m_2 \frac{d\vec{v}_2}{dt}$$

$$\Rightarrow m_1 \vec{a}_1 = -m_2 \vec{a}_2 \Rightarrow \frac{m_2}{m_1} = -\frac{\vec{a}_1}{\vec{a}_2}$$

• We can always choose one of the masses as the unit mass, and we can compare the accelerations to ~~derive~~ measure the mass of the other body.

$m_1 = \text{unit mass} \stackrel{\text{assume that}}{=} 1 \text{ kg}$

$$m_2 = -\frac{a_1}{a_2} \text{ kg}$$

There is another way of measuring mass: comparing the weight of 1 body with a standard unit.

$$\vec{F} = m\vec{a} \rightarrow \vec{W} = m\vec{g}$$

$$\frac{W_1}{W_2} = \frac{m_1}{m_2} \frac{g_1}{g_2} \quad ^0$$

Lecture 2.1 : Newton's "laws" of motion

- Physical laws must be based on experimental fact.
- We come up with a postulate which refer to the experimental data various implications by this postulate can be made.
- If these implications are verified by the experiment the postulate gains the status of a "physical law".

Newton's laws of mechanics : (no change in the velocity)

- 1) A body remains at rest or in "uniform motion" unless acted upon by a force.
- 2) A body acted upon a force moves in such a manner that the time rate of change of its "momentum" equals the force.
- 3) If two bodies exert forces on each other, these forces are equal in magnitude and opposite in direction.

Explanation of the "laws":

1) → Defines what zero force does.

2) → Defines what nonzero force does: It also involves "first law".

$$\frac{d}{dt} [\text{momentum}] = 0 \rightarrow \text{first law.}$$

★ 1st and 2nd statements actually defines what force and momentum mean. Without these statements we have no knowledge what to do with force and momentum qualitatively. Therefore, 1st and 2nd statements are actually definitions rather than laws.

★ If the 2nd law involves the 1st already, why do we need the 1st law? → It defines "an inertial reference frame".

→ e.g. what do we feel in a car taking a turn?

★ what is momentum? $\vec{p} \equiv m \vec{v} \Rightarrow$ 2nd law: $\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt} [m \vec{v}]$

Therefore, 1st & 2nd statements are actually the definitions of Force and mass (and consequently momentum)

$$\rightarrow \text{If } m \text{ is constant} \rightarrow \vec{F} = m \frac{d\vec{v}}{dt} = \boxed{\vec{F} = m \vec{a}}$$

★ 3rd law is indeed a law. It is a statement concerning the real world and confirms all the physics in Newton's laws of motions.

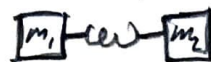
★ More about 3rd law $\rightarrow$ It is not universal! ▽

It only applies when the force exerted by one (point) object on another (point) object is directed along the line connecting the objects. $\rightarrow$ CENTRAL FORCES ▽ e.g. Gravitational & electrostatic forces.

$$\vec{F}_g = G \frac{m_1 m_2}{r^2}$$

$$\vec{F}_e = k \frac{q_1 q_2}{r^2}$$

two objects connected by a spring



NON CENTRAL FORCES: Velocity dependent forces
e.g. force b/w moving electric charges

★ We have said that 1st and 2nd statements defines the mass and force. So, how are we going to measure the mass qualitatively?

$\rightarrow$ 3rd law can help us out! How?

Let's state the 3rd law with different words:

3rd law: If 2 bodies form an ideal isolated system, then the accelerations of them are always in opposite directions (and the ratio of the magnitudes of the accelerations is constant. This constant ratio is the inverse ratio of the masses of the bodies)

Start with



Def of iso syst

$$\sum \vec{F}_i = 0$$

$$\vec{F}_1 = -\vec{F}_2 \rightarrow \frac{d[\vec{p}_1]}{dt} = -\frac{d[\vec{p}_2]}{dt}$$

two $\vec{a}$ vectors are oppositely directed.

$$m_1 \frac{d[\vec{v}_1]}{dt} = -m_2 \frac{d[\vec{v}_2]}{dt} \rightarrow m_1 \vec{a}_1 = -m_2 \vec{a}_2 \rightarrow \frac{m_2}{m_1} = -\frac{\vec{a}_1}{\vec{a}_2}$$

we can always select one of the masses as the unit mass and by comparing the $\vec{a}$'s we can determine the mass of other body. To measure the $\vec{a}$'s we need clocks and measuring rods and a "suitable ref. frame", i.e., an inertial ref. frame

★ Another way of measuring is weighing: comparing the weight of 1 body with a standard (unit).

This procedure makes use of in a gravitational field the weight of a body is just the gravitational force acting on the body

$$\vec{F} = m\vec{a} \rightarrow \vec{W} = m'\vec{g}$$

$$\frac{W_1}{W_2} = \frac{m_1 \left(\frac{g_1}{g_2}\right)}{m_2 \left(\frac{g_1}{g_2}\right)}$$

$\rightarrow$ we know it by experiment.

m : inertial mass

m' : gravitational mass

} experiment shows with a very high precision that these two masses are the same.

Conservation of the linear momentum: In an isolated system:

$$\frac{d}{dt}[\vec{p}_1] = -\frac{d}{dt}[\vec{p}_2] \rightarrow \frac{d}{dt}[\vec{p}_1 + \vec{p}_2] = 0 \Rightarrow \vec{p}_1 + \vec{p}_2 = \text{constant}$$

Galilean invariance or principle of Newtonian relativity

If Newton's laws are valid in one ^{inertial} reference frame, they are also valid in any reference frame in uniform motion with respect to the first system: $\vec{F} = m\vec{\ddot{x}}$ involves the second der. of $\vec{x}$; a change of coordinates involving a constant velocity does not influence the equation.

eg. S and S'

Absolute time: If two inertial frames fix the units (seconds, hours...) in which to measure the duration time, then the only remaining choice they can make is when to start clock; time flows exactly in the same way in both frames.

$$t' = t + t_0 \rightarrow \text{absolute tm.}$$

Lecture 2.2: The Equation of Motion for a Particle.

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}[m\vec{v}] = m \frac{d}{dt}[\vec{v}] = m\vec{\ddot{r}} \Rightarrow \vec{F} = m\vec{\ddot{r}}$$

constant mass assumption

Equation of motion

$$= m \frac{d^2}{dt^2}[\vec{r}(t)] : \text{second-order diff eq.}$$

We want to find $\vec{r}(t)$; ^(the trajectory) how particle moves. This can be done by integrating $\vec{F} = m \frac{d^2}{dt^2}[\vec{r}(t)]$ twice, once we know what $\vec{F}$ is and what is the initial values of the position and the velocity of the particle.

Example: A particle accelerates in 1D with a constant force F . Solve its equation of motion for the ini. cond. $v(0) = v_0$; $x(0) = x_0$

$$\vec{a}, \vec{v} \rightarrow x \rightsquigarrow x(t) = ? \quad v(t) = ?$$

$$\vec{F}_{\text{net}} = m\vec{\ddot{r}} \rightarrow F_x = m\ddot{x} \rightarrow \frac{F_x}{m} = \frac{d}{dt}\dot{x} \rightarrow \frac{F_x}{m} dt = d\dot{x}$$

$$\rightarrow \int d\dot{x} = \frac{F_x}{m} \int dt + C_1 \rightarrow \dot{x}(t) = v(t) = \frac{F_x}{m} t + C_1 ; \text{what is } C_1 ?$$

$$v(t=0) = C_1 = v_0 \Rightarrow v(t) = \frac{F_x}{m} t + v_0$$

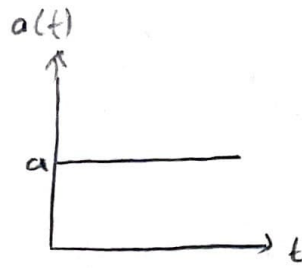
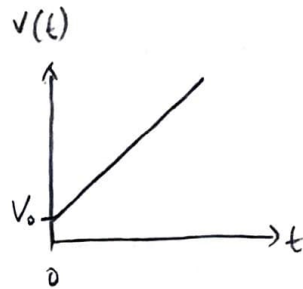
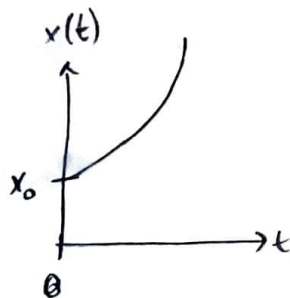
$$v(t) = \frac{d}{dt}[x(t)] = \frac{F_x}{m} t + v_0 \rightarrow d[x(t)] = \frac{F_x}{m} t dt + v_0 dt \rightarrow \int d[x(t)] = \frac{F_x}{m} \int t dt + v_0 \int dt + C_2$$

$$\rightarrow x(t) = \frac{F_x}{2m} t^2 + v_0 t + C_2 \rightarrow x(t=0) = x_0 \rightarrow \frac{F_x t^2}{2m} + v_0 t + x_0$$

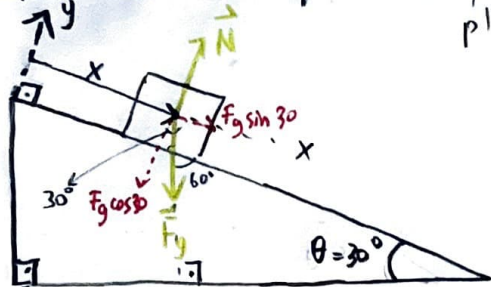
$$x(t) = \frac{1}{2} \frac{F}{m} t^2 + v_0 t + x_0$$

$$v(t) = \frac{F}{m} t + v_0$$

$$a(t) = a$$



Example: Inclined plane: If a block slides without friction down a fixed, inclined plane with $\theta = 30^\circ$ what is the block's acceleration?



$\vec{F}_{\text{net}} = \vec{F}_g + \vec{N}$: This does not change throughout the motion.

$$\vec{F}_{\text{net}} = m\vec{a} \rightarrow \vec{F}_g + \vec{N} = m\vec{a}$$

Define the simplest reference frame explaining the motion: x-y

y-direction: $F_g \cos \theta + N = 0$: We know that the block does not move in y direction

x-direction: $F_g \sin \theta = m\ddot{x} \Rightarrow \ddot{x} = \frac{F_g \sin \theta}{m} = \frac{mg \sin \theta}{m} = g \sin \theta \Rightarrow \ddot{x} = \frac{9.8}{2} \frac{m}{s^2} = 4.9 \frac{m}{s^2}$
 constant

Retarding Forces: The force which stops the body or tends to stop the body during the motion: $\vec{F} \propto -\vec{v}$

e.g. Friction as we will see in the exercise hour.

We will examine one more retarding force: drag

Drag: The force of resistance to motion through a fluid.

e.g. Submarines, planes, air friction for cars...

Drag forces are usually proportional to some power of speed. In general, the real drag forces are more complicated, but the power-law approximation is useful in many instances in which the speed does not vary greatly.

$$F_d \propto v^r \text{ drag force}$$

Example: Free fall and terminal speed

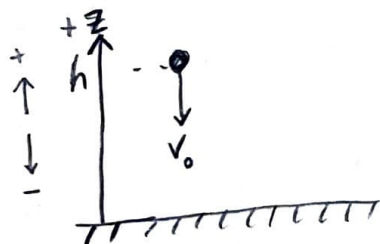
Find the displacement and velocity of a particle undergoing vertical motion in a medium having a retarding force proportional to velocity.

Assume h : initial height
 v_0 : initial velocity.
 (in the gravitational field)

h is decreasing in time $\Rightarrow v = \dot{h} < 0$ by definition.
 $\Rightarrow F_d = kv < 0$ for all t . If we take $g = 9.8 \text{ m/s}^2 > 0$ then $mg > 0$. To write the net force, we need to take care of these quantities carefully. kv is towards upwards and mg is towards downwards.

$$F_d(t) = kV(t)$$

drag coefficient



$$mg + kv = F_{\text{net}}$$

$$-mg - kv = F_{\text{net}}$$

after this we will only consider magnitudes

$$-mg - kv = F_{\text{net}} = ma = m \frac{dv}{dt}$$

$$\frac{dv}{dt} = -g - \frac{k}{m}v \Rightarrow \frac{dv}{-g - \frac{k}{m}v} = -dt$$

Substitute: $g + \frac{k}{m} v = l \Rightarrow \frac{dl}{dv} = \frac{k}{m} \Rightarrow dv = \frac{m}{k} dl$

$$\Rightarrow \frac{1}{l} \frac{m}{k} dl = -dt \Rightarrow \frac{m}{k} \int \frac{dl}{l} = -\int dt + C \Rightarrow \frac{m}{k} \ln l = -t + C$$

$$\Rightarrow \frac{m}{k} \ln \left[g + \frac{k}{m} v \right] = -t + C \Rightarrow \ln \left[g + \frac{k}{m} v \right] = -\frac{k}{m} t + \frac{k}{m} C$$

Initial condition:

$t=0 \Rightarrow v(t) = V_0 \Rightarrow \ln [g + \frac{k}{m} V_0] = \frac{k}{m} C$ } skip this for now
insert it here;

Also substitute:

$$\frac{k}{m} = \omega^2$$

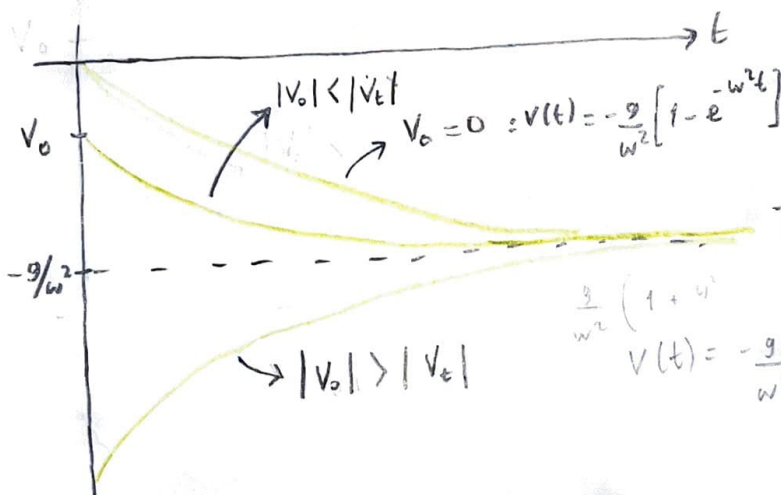
$$g + \omega^2 v = e^{-\omega^2 t} e^{+\omega^2 C} \Rightarrow g + \omega^2 v = e^{-\omega^2 t} [g + \omega^2 V_0]$$

$$\Rightarrow v(t) = -\frac{g}{\omega^2} + \frac{g + \omega^2 V_0}{\omega^2} e^{-\omega^2 t}$$

$$v(0) = -\frac{g}{\omega^2} + \frac{g}{\omega^2} + \frac{\omega^2 V_0}{\omega^2}$$

when $t \rightarrow \infty$, $e^{-\omega^2 t} \rightarrow 0$

$$\Rightarrow v(t \rightarrow \infty) = -g/\omega^2$$



Terminal speed $|v(t \rightarrow \infty)| = g/\omega^2 = |V_t|$

$$v(t) = -\frac{g}{\omega^2} + \frac{g}{\omega^2} e^{-\omega^2 t} + V_0 e^{-\omega^2 t}$$

$$z(t) = -\frac{g}{\omega^2} \int dt + \frac{g + \omega^2 V_0}{\omega^2} \int e^{-\omega^2 t} dt + C$$

$$= -\frac{gt}{\omega^2} - \left[\frac{g + \omega^2 V_0}{\omega^2} \right] \frac{1}{\omega^2} e^{-\omega^2 t} + C$$

$$z(0) = -\frac{g + \omega^2 V_0}{\omega^2} \frac{1}{\omega^2} + C = h \Rightarrow C = h + \frac{g + \omega^2 V_0}{\omega^4}$$

$$\Rightarrow z(t) = -\frac{gt}{\omega^2} - \frac{g + \omega^2 V_0}{\omega^4} e^{-\omega^2 t} + h + \frac{g + \omega^2 V_0}{\omega^4}$$

$$= h - \frac{gt}{\omega^2} + \frac{\omega^2 V_0 + g}{\omega^4} [1 - e^{-\omega^2 t}]$$